Devansh Singh Bais

+91-8770019678 | devanshsinghbais111@gmail.com | github.com/DevanshSinghBais | linkedin.com/in/devansh-singh-bais-748b192b5

Education

Maulana Azad National Institute of Technology, Bhopal — Bhopal, India
B.Tech. in Computer Science and Engineering; CGPA: 9.15/10 — *2025 – Apr. 2029*

Garima Vidya Vihar Senior Secondary School — Indore, India
CBSE Class XII; 90.5% — *2025*

PM SHRI Kendriya Vidyalaya No. 2 — Indore, India
CBSE Class X; 94.5% — *2023*

Research & Technical Experience

BCI Club, Maulana Azad National Institute of Technology — Bhopal, India
Research & Technical Executive — *2025 – Present*

- Conduct research and technical exploration across **Artificial Intelligence, Machine Learning, and Brain-Computer Interfaces (BCI)**, with emphasis on translating research concepts into practical implementations.
- Read and analyze research papers, investigate experimental AI/ML problems, and contribute to technical discussions involving **neural networks and NLP**.
- Contribute to technical projects, workshops, research discussions, and technical events while collaborating with multidisciplinary student teams.
- Mentor peers on technical projects and contribute to project planning, implementation, coordination, and team execution.

Projects

SafeHer – Women's Safety Platform | *TypeScript, PostgreSQL, Web APIs, Location Services* — 2025

- Developed a women's safety platform providing **real-time assistance, emergency support, location-based services, and rapid alert workflows**.
- Integrated application services and external APIs to support safety-focused workflows, location-aware functionality, and emergency assistance features.
- Collaborated in a hackathon team to design and implement a real-world social-impact solution under strict time constraints.
- **Won 1st Prize at the BITS Pilani Webathon 2025** for the SafeHer project.

StudyPilot – AI Study & Research Agent | *Python, LLMs, Gradio, Tool Calling, APIs* — 2026

- Developed an AI-powered study and research assistant integrating **LLM reasoning, tool calling, local knowledge, and interactive AI workflows**.
- Integrated multiple external services and AI APIs into modular workflows, contributing to experience across **25+ API integrations**.
- Built an interactive **Gradio** interface for conversational study, research, and knowledge-assistance workflows.
- Designed extensible AI workflows connecting language models with external tools and knowledge sources.

GitHub Repo Explainer – LLM Codebase Analyzer | *Python, Gemini API, GitHub REST API* — 2026

- Built an LLM-powered developer tool for analyzing GitHub repositories and generating contextual explanations of codebases and project structure.
- Integrated the **GitHub REST API** with Google's Gemini models to retrieve repository information and perform AI-assisted code analysis.
- Developed API-driven workflows for processing repository data and generating structured, context-aware explanations using LLMs.
- Applied LLM-based developer tooling to simplify understanding and exploration of unfamiliar software repositories.

Machine Learning & Neural Networks Implementations | *Python, NumPy, Matplotlib, Scikit-learn* — 2026

- Implemented core machine learning algorithms and optimization techniques including **linear regression, logistic regression, gradient descent, and regularization**.
- Implemented and experimented with neural networks to develop practical understanding of forward propagation, loss computation, and gradient-based optimization.
- Used **NumPy** for numerical computation and **Matplotlib** for visualizing model behavior, optimization, and regularization effects.

Achievements

- **1st Prize – BITS Pilani Webathon 2025** for SafeHer, a women's safety platform.
- **99.48 percentile** in JEE Main; All India Rank: **9,123**.
- **Gold Medal – School Level**, International Mathematics Olympiad (IMO).
- **Gold Medal – School Level**, International Science Olympiad (ISO).

Competitive Programming

- **LeetCode:** 300+ problems solved; rating **1550**.
- **Codeforces:** 200+ problems solved; rating **1400**.
- **CodeChef:** 2-Star rated programmer.
- Solved **500+ algorithmic problems** across competitive programming platforms, strengthening data structures, algorithms, and problem-solving skills.

Technical Skills

Languages: Python, C, C++, JavaScript, TypeScript, SQL, HTML, CSS
AI / Machine Learning: NumPy, Pandas, Matplotlib, Scikit-learn, Neural Networks, Gradient Descent, Regularization, Machine Learning
Generative AI: OpenAI API, Google Gemini API, Anthropic API, Hugging Face Transformers, Hugging Face Pipelines, LangChain, LangGraph, RAG, Tool/Function Calling, Gradio
Web / Backend: React.js, Node.js, Express.js, MongoDB, PostgreSQL, MySQL, Flask, FastAPI, REST APIs
Developer Tools: Git, GitHub, Docker, Linux, Jupyter Notebook, Google Colab, VS Code, Cursor
API Engineering: 25+ Third-Party API Integrations, REST API Development, API Authentication, API Orchestration

Certifications & Courses

- **Machine Learning Specialization** – Andrew Ng, Coursera — Certificate
- **LLM Engineering** – Ed Donner, Udemy — Certificate
- **MERN Stack Development** – Angela Yu, Udemy
- **Neural Networks: Zero to Hero** – Andrej Karpathy